

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

Assignment No.: 24

Date: 08/04/2026

Lab Assignment: 24

Aim:

To write a program to access a file after it is closed.

Objective:

- To understand the behavior of files after closing them in Python.
- To learn the importance of properly managing file operations.
- To observe the error generated when trying to access a closed file.
- To reinforce good programming practices in file handling.

Theory:

In Python, file handling is performed using the `open()` function, which returns a file object. After performing operations like reading or writing, it is important to close the file using the `close()` method. Once a file is closed, it cannot be accessed again using the same file object. Any attempt to perform operations such as reading or writing on a closed file will result in an error (specifically, a `ValueError`). This behavior ensures proper memory management and prevents accidental data corruption. Therefore, programmers must ensure that all file operations are completed before closing the file, or reopen the file if further access is required.

Algorithm:

1. Start the program.
2. Specify the file name.
3. Open the file in read mode.
4. Display a message indicating the file is opened.
5. Close the file using the `close()` method.
6. Display a message indicating the file is closed.
7. Try to read or access the file after closing it.
8. Observe and display the error generated.
9. End the program.

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Pseudo Code:

```
START
OPEN file in read mode
PRINT "File opened"
CLOSE file
PRINT "File closed"
TRY to READ file
DISPLAY error message
END
```

Code:

```
# Open the file
file = open("Sample.txt", "r")

# Close the file
file.close()

# Try to read after closing
try:
    content = file.read()
    print(content)
except ValueError as e:
    print("Error:", e)
```

Output:

```
Error: I/O operation on closed file.
```

```
[Done] exited with code=0 in 0.586 seconds
```

Conclusion:

Thus, the Python program was successfully executed to demonstrate that a file cannot be accessed after it is closed. Attempting to perform operations on a closed file results in an error, highlighting the importance of proper file handling and management in Python.